

Now, Minsky proposes a different test that would determine when machines have reached a level of sophistication that could begin to be truly useful. His method? To test whether the machine can read a simple children's book, understand what the story is about, and explain it in its own words or ask reasonable questions about it.

It's not clear whether that's an achievable goal on this kind of timescale, but Gershenfeld says, "We need good challenging projects that force us to bring our program together."

A co-processor for the human brain

Of course, as with all talks about artificial intelligence, there comes a time when you have to breach the topic of hybrid human-artificial minds, or "bionic brains". While the scientists don't want to speculate about that, they are working on something that they like to call a "co-processor for the brain".

This system, also referred to as a cognitive assistive system, would initially be aimed at people suffering from cognitive disorders such as Alzheimer's disease. The concept is that it would monitor people's activities and brain functions, determine when they needed help, and provide exactly the right bit of helpful information at the right time. For example, with a faulty memory in the patient, the co-processor could tell the brain the name of a person who just entered the room, and information about when the patient last saw that person.

The same kind of system could also find applications for people without any disability, as a form of brain augmentation – a way to enhance their own abilities, for example by making everything from personal databases of information to all the resources of the internet instantly available just when it's needed. The idea is to make the device as non-invasive and unobtrusive as possible – perhaps something people would simply slip on like a pair of headphones.

In the longer run, Minsky still sees hope for far grander goals. For example, he points to the fact that his iPhone can now download thousands of different applications, instantly allowing it to perform new functions. Why not do the same with the brain? "I would like to be able to download the ability to juggle," he says. "There's nothing more boring than learning to juggle." Is the Matrix real? 